

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S1 (Special Improvement) Examination January 2021 (2019 scheme)

Course Code: MAT101**Course Name: LINEAR ALGEBRA AND CALCULUS**
(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

- 1 Determine the rank of the matrix $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix}$ (3)
- 2 What kind of conic section is represented by the quadratic form $7x_1^2 + 6x_1x_2 + 7x_2^2 = 200$ Transform it into canonical form. (3)
- 3 Find the derivative of $w = x^2 + y^2$ with respect to t along the path $x = at^2, y = 2at$. (3)
- 4 Let $f(x, y) = \sqrt{3x + 2y}$, find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(2, 5)$. (3)
- 5 Evaluate $\int_0^a \int_0^a \int_0^a (yz + xz + xy) dx dy dz$ (3)
- 6 Use polar co-ordinates to evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{\frac{3}{2}} dy dx$ (3)
- 7 Test the convergence of $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$ (3)
- 8 Examine whether the series convergence or not $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ (3)
- 9 Find the Maclaurin series of $\frac{1}{x+1}$ up to third degree term. (3)
- 10 Find the Fourier Half Range sine series of $f(x) = x$ in $, 0 < x < \pi$. (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module-I**

- 11 a) Test for consistency and solve the system of equations (7)

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

$$x - y + z = -1$$

- b) Find the eigenvalues and eigenvectors of $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ (7)

- 12 a) For what values of a and b do the system of equations (7)

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + az = b$$

have i) no solution ii) unique solution iii) more than one solution.

- b) Find the matrix of transformation that diagonalize the matrix (7)

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \text{ Also write the diagonal matrix.}$$

Module-II

- 13 a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$. (7)

- b) If the local linear approximation of a function $f(x, y, z) = xy + z^2$ at a point P is $L(x, y, z) = y + 2z - x$, find the point P. (7)

- 14 a) If $z = e^{xy}$, $x = 2u + v$, $y = \frac{u}{v}$ find $\frac{\partial z}{\partial u}$. (7)

- b) Locate all relative extrema of $f(x, y) = 3x^2 - 2xy + y^2 - 8y$. (7)

Module-III

- 15 a) Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by reversing the order of integration. (7)

- b) Using triple integral find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x + 2y + z = 6$. (7)

0000MAT101121904

- 16 a) Find the mass and center of gravity of the triangular lamina with vertices (0,0), (0,1) and (1,0) and density function $\delta(x,y) = xy$ (7)

- b) Evaluate $\iint_R x^2 dy dx$, where R is the region between $y = x$ and $y = x^2$ (7)

Module-IV

- 17 a) Discuss the convergence of the series (7)

$$(i) \sum_{k=1}^{\infty} \frac{k!}{k^k} \quad (ii) \sum_{k=1}^{\infty} \left(\frac{k}{k+1} \right)^{k^2}$$

- b) Examine the convergence and divergence of the series (7)

$$\frac{x}{1.3} + \frac{x^2}{3.5} + \frac{x^3}{5.7} + \dots$$

- 18 a) Test the convergence of $1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \frac{1.3.5.7}{7!} + \dots$ (7)

- b) Prove that the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{(k-2)}{k(k+1)}$ is conditionally convergent. (7)

Module-V

- 19 a) Obtain Fourier series for the function $f(x) = |\sin x|$ $-\pi < x < \pi$ (7)

- b) If $f(x) = \begin{cases} kx ; 0 < x < \frac{\pi}{2} \\ k(\pi - x) ; \frac{\pi}{2} < x < \pi \end{cases}$ then show that (7)

$$f(x) = \frac{4k}{\pi} \left(\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \dots \right).$$

- 20 a) Find the Fourier cosine series of $f(x) = x^2$ in $(0, \pi)$. Hence show that (7)

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

- b) Find the Fourier series for the function (7)

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 1 - x & 1 < x < 2 \end{cases}$$
